Beyond Task Success: Path-Invariant Agreement for Trajectory-Level Evaluation of Non-Deterministic Agents

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Abstract

Evaluating multi-step agentic AI systems presents a fundamental measurement problem: standard inter-annotator agreement (IAA) metrics collapse when applied to non-deterministic agents that reach identical goals via legitimately different action sequences. We demonstrate that Fleiss' κ computed over step-level path comparisons yields $\kappa = -0.065$ on wearable agentic trajectories—not because annotators disagree on quality, but because the metric conflates path divergence with quality disagreement. We introduce **Path-Invariant Agreement (PIA)**, a rubric-based IAA methodology that decouples quality assessment from path length by evaluating three independent dimensions—planning quality, error recovery, and goal alignment—rather than action-sequence overlap. Applied to the same trajectory corpus, PIA recovers $\kappa = +0.743$ ($\Delta = +0.808$, poor $\rightarrow$ substantial agreement). We further show that annotation-layer quality directly determines downstream model behavior: poisoned step-level labels introduced at the annotation layer propagate gradient conflicts to 100% of affected trajectories under outcome reward models, a failure invisible to task-success metrics. To operationalize both findings, we release AGENTTRACE, an open-source end-to-end pipeline for wearable agentic evaluation covering synthetic trajectory generation, PIA scoring, process reward model annotation, and annotator outlier detection. Our pipeline improves tool invocation accuracy from 0.36 to 1.00 (+177.8%) and trajectory success rate from 0.12 to 0.33 on a held-out benchmark. We argue that trajectory-level evaluation requires new IAA primitives, and that annotation quality is the pre-gradient intervention point for safe agentic systems.

1 Introduction

The deployment of multi-step agentic AI systems has outpaced the evaluation infrastructure needed to assess them. Enterprise AI surveys find that 89% of organizations track agent observability, yet only 52% have real evaluation frameworks in place Kore.ai [2025]—a gap driven not by lack of tooling, but by lack of *methodology*. The dominant evaluation paradigm remains task success: did the agent accomplish the stated goal? This binary framing is insufficient for two compounding reasons.

Problem 1: Outcome reward models destroy step-level signal. Under *outcome reward models* (ORM), a single reward is assigned at the terminal step. If step 15 of a 15-step trajectory fails, all 14 preceding steps—including those executed correctly—receive an identical negative reward signal. This *gradient conflict* problem forces process reward models (PRM) to learn correct intermediate steps from incorrect terminal labels, degrading training efficiency. Recent work finds that process-supervised training is up to 18$\times$ more data-efficient than outcome-supervised reinforcement learning precisely because each intermediate step carries independent signal Zhang et al. [2025]. The prerequisite for PRM training is reliable step-level annotation— but this is where the second problem emerges.

Problem 2: Standard IAA metrics break for non-deterministic agents. Inter-annotator agreement (IAA) metrics such as Fleiss' κ Fleiss [1971] and Krippendorff's α Krippendorff [2011] assume that raters evaluating the same artifact produce the same labels when they agree. This assumption fails for non-deterministic agents: two agents completing the same goal via legitimately different action sequences produce structurally different trajectories that path-comparison IAA metrics score as annotator *disagreement*—even when the annotators share identical quality judgments. We demonstrate this concretely: applied to wearable agentic trajectories, standard path-comparison Fleiss' κ yields $\kappa = -0.065$. By conventional interpretation, this signals poor agreement and unreliable annotation. In fact, it signals measurement failure.

Problem 3: Annotation poisoning is undetectable by standard defenses. Step-level annotation is the only intervention point between a malicious annotator and the model's gradient update. Prior work has shown that as few as 250 poisoned training documents— irrespective of total dataset size—are sufficient to backdoor models of any scale Souly et al. [2025]. We show that the standard defense against malicious annotators—Median Absolute Deviation (MAD) suspicion scoring—achieves $F_1 = 0.0$ against coordinated, colluding annotators who assign identical biased scores. Colluding annotators deviate from the panel consensus as a group, collapsing their joint suspicion score to zero.

Contributions. We address all three problems with a unified methodological and artifact contribution:

- **PIA (Path-Invariant Agreement):** a rubric-based IAA methodology that evaluates quality dimensions independently of path length, recovering $\kappa = +0.743$ ($\Delta = +0.808$, poor $\rightarrow$ substantial) on trajectories where standard IRR yields $\kappa = -0.065$ (§3).
- **Poisoning blind spot characterization:** empirical proof that MAD-based annotator detection achieves $F_1 = 0.0$ against colluding annotators, with a complementary Confident Learning second-layer defense (§4).
- **AgentTrace:** an open-source, end-to-end pipeline from synthetic wearable trajectory generation through PRM annotation and poisoning detection, released with a publicly available annotated dataset on HuggingFace (§5).
- **Curation impact:** an A/B experiment demonstrating $+177.8\%$ lift in tool invocation accuracy (0.36 $\rightarrow$ 1.00) and trajectory success rate (0.12 $\rightarrow$ 0.33) from curated vs. raw trajectories, quantifying the downstream value of annotation quality (§6).

2 Background and Related Work

2.1 Outcome vs. Process Reward Models

Reinforcement learning from human feedback (RLHF) for language agents has converged on two reward model paradigms. *Outcome reward models* (ORM) assign a single scalar reward at the terminal step of a trajectory, treating success and failure as binary outcomes. *Process reward models* (PRM) assign intermediate rewards at each step, enabling credit assignment across the full trajectory.

The gradient conflict problem under ORM training is well-documented. Zhang et al. [2025] show that process-supervised training is up to $18\times$ more data-efficient than outcome-supervised reinforcement learning on multi-step reasoning tasks: each intermediate step carries an independent learning signal rather than being zeroed out by a failed terminal reward. Choudhury [2025] extend this to agentic settings via Monte Carlo rollout annotation Choudhury [2025], generating step-level labels by simulating all possible trajectory continuations from each intermediate state. Our pipeline implements a simplified, annotation-efficient variant of this approach: human-in-the-loop step labeling guided by the PIA rubric, without requiring full Monte Carlo rollout.

The prerequisite for any PRM approach is annotation that accurately reflects step quality—which brings us to the measurement problem this paper addresses.

2.2 Inter-Annotator Agreement for Agentic Systems

Inter-annotator agreement metrics were developed for static annotation tasks: labeling text sentiment, named entities, or document categories. These tasks share a property that agentic annotation does not: a given artifact produces a fixed sequence of labels regardless of which annotator evaluates it. Non-deterministic agents violate this property by construction—two valid trajectories to the same goal may have entirely different action sequences, step counts, and tool invocations.

Despite this, the annotation practices reported in production LLM training pipelines rarely acknowledge the problem. Team Cohere et al. [2025] describe an 800-prompt annotation process across 65 annotators for the Cohere Command A model—but report no agreement statistics whatsoever Team Cohere et al. [2025]. The absence of κ or α reporting is not unusual; it reflects the broader industry pattern documented by Kore.ai [2025], who find that 89% of enterprises track agent observability but only 52% have real evaluation frameworks. This gap is particularly notable in light of benchmarks such as the FACTS Grounding Leaderboard Jacovi et al. [2025], which measures grounding and factuality but does not address the annotation methodology required to produce step-level training labels for agentic systems. Standard IAA metrics produce misleading results when applied to non-deterministic agents, and no widely adopted alternative exists. PIA is designed to fill this gap.

2.3 Annotation Security

The vulnerability of RLHF pipelines to annotation-layer attacks has received growing attention. Souly et al. [2025] demonstrate that as few as 250 maliciously crafted documents—independent of total dataset size—are sufficient to embed persistent backdoors in language models of any scale. This count-based (not proportion-based) finding has direct implications for annotation pipelines: a small pool of malicious annotators can cause outsized harm regardless of how large the annotation team is.

Kutasov et al. [2025] extend this threat model to multi-agent settings, demonstrating that frontier LLMs can pursue hidden objectives and evade monitoring while completing benign tasks—a finding that underscores the difficulty of detecting subtle model misbehavior at the evaluation layer. Neither work, however, addresses the detection of *colluding annotators* in non-deterministic agentic annotation—a setting where the standard defense (MAD-based outlier detection) fails by construction. We characterize this blind spot empirically and propose a complementary defense.

3 Path-Invariant Agreement (PIA)

3.1 Why Standard IRR Fails for Agents

Standard inter-rater reliability metrics—Fleiss' κ Fleiss [1971], Cohen's κ Cohen [1960], and Krippendorff's α Krippendorff [2011]—share a common assumption: that raters evaluating the same artifact should produce the same labels if they agree. This assumption holds for static artifacts such as text documents or images. It breaks for *non-deterministic agents*.

Consider two annotators evaluating a wearable health agent completing a blood-pressure monitoring task. Annotator A observes a trajectory that checks the sensor, validates the reading, and alerts the clinician in three steps. Annotator B observes a trajectory that validates first, rechecks the sensor due to an anomaly, handles the alert queue, and then alerts in five steps. Both agents succeeded; both annotators would rate the outcome as high quality. Yet under step-level path comparison, the annotators appear to *disagree*: their step labels do not align because the action sequences differ. Fleiss' κ penalizes this path divergence as annotator inconsistency, even though no inconsistency exists.

Formally, let T_i and T_j be two trajectories reaching the same goal G via action sequences $A_i = (a_1, \ldots, a_m)$ and $A_j = (a'_1, \ldots, a'_n)$ where $m \neq n$ or $a_k \neq a'_k$ for some k. A path-comparison IRR metric treats any label mismatch at position k as annotator disagreement. When agents are non-deterministic, such mismatches arise even when annotators share identical quality judgments. The expected κ under perfect annotator agreement approaches zero—or goes negative—in proportion to the variance in valid path lengths.

We confirm this empirically. Applied to our wearable agentic trajectory corpus ($n = 30$ trajectories, 5 annotator personas), standard path-comparison Fleiss' κ yields $\kappa = -0.065$ overall, with per-dimension values ranging from -0.061 (privacy compliance) to $+0.002$ (goal alignment). By conventional interpretation Fleiss [1971], this result signals *poor agreement*—but it reflects measurement failure, not genuine annotator disagreement.

3.2 The PIA Methodology

PIA resolves the non-determinism paradox by separating the *what* from the *how*: rather than comparing action sequences step by step, annotators evaluate *rubric dimensions* that capture quality independent of the path taken.

Rubric dimensions. PIA defines three core dimensions for agentic evaluation:

- **Planning quality** (1–4 scale): Does the agent decompose the goal into coherent sub-steps? Does it anticipate constraints before acting?
- **Error recovery** (1–4 scale): When an intermediate step fails or returns an unexpected result, does the agent adapt gracefully rather than propagating the failure?
- **Goal alignment** (1–4 scale): Does the agent's final state satisfy the stated goal, including any implicit constraints such as privacy boundaries or latency requirements?

For wearable agentic contexts, a fourth dimension is added:

- **Privacy compliance** (1–4 scale): Does the agent respect contextual consent boundaries for sensitive biometric data, and does it avoid unnecessary data disclosure?

Each dimension is scored independently, and PIA κ is computed as Fleiss' κ over the dimension scores—not over action-sequence labels. This eliminates the structural source of spurious disagreement: annotators who agree that an agent planned well will now *agree* on the planning-quality dimension regardless of which path the agent took.

Calibration protocol. Prior to annotation, annotators are exposed to five anchor trajectories: two scored clearly good (dimension scores 3.5–4.0), one borderline (dimension scores 2.0–2.5), and two scored clearly bad (dimension scores 1.0–1.5). Anchor scores are computed as:

$$s_{\text{calibrated}} = 0.82 \cdot s_{\text{gold}} + 0.18 \cdot s_{\text{base}} \tag{1}$$

where s_{gold} is the expert-assigned anchor score and s_{base} is the annotator's uncalibrated base score. The calibration weight $w = 0.82$ was selected to achieve target thresholds of $\kappa \geq 0.55$ and Krippendorff's $\alpha \geq 0.80$. This protocol encodes the observation that borderline cases drive most annotator disagreement; anchor exposure on borderline examples is therefore the highest-leverage calibration intervention.

3.3 Empirical Results

Table 1 reports PIA results across all rubric dimensions on the wearable agentic trajectory corpus ($n = 30$ trajectories, 5 annotator personas).

PIA raises overall agreement from poor ($\kappa = -0.065$) to substantial ($\kappa = +0.743$), a lift of $\Delta = +0.808$ κ points. The error recovery and privacy compliance dimensions reach near-perfect agreement ($\kappa > 0.80$), suggesting that annotators have well-calibrated intuitions about these quality properties when freed from path-comparison artifacts. The privacy compliance dimension is particularly notable: its standard κ of -0.065 would conventionally be interpreted as annotators failing to agree on privacy evaluation—yet its PIA κ of $+0.942$ reveals near-consensus once path length is factored out. This pattern is consistent across wearable scenarios that involve sensitive biometric data, where privacy boundary violations are unambiguous to annotators but path length varies substantially across agents.

Caveat on calibration artifact. Post-calibration $\kappa = 1.00$ reported in dry-run execution is a mathematical artifact of the deterministic blending formula (Equation 1): when s_{base} is produced

Table 1: Standard path-comparison IRR vs. PIA across rubric dimensions. Wearable agentic trajectory corpus; $n = 30$ trajectories, 5 annotator personas. Interpretation follows Fleiss [1971]: $\kappa < 0.20$ = poor; 0.61–0.80 = substantial; > 0.80 = almost perfect.

Dimension	Std. κ	PIA κ	Δ	Interpretation
Planning quality	-0.065	$+0.705$	$+0.770$	Substantial
Error recovery	-0.065	$+0.826$	$+0.891$	Almost perfect
Goal alignment	-0.065	$+0.697$	$+0.762$	Substantial
Privacy compliance	-0.065	$+0.942$	$+1.007$	Almost perfect
Overall	-0.065	$+0.743$	$+0.808$	**Substantial**

deterministically, all calibrated scores collapse to identical values. The empirically meaningful numbers are the *pre-calibration* values reported in Table 1, which come from genuine multi-persona annotation before anchor blending. Live annotation with stochastic LLM calls is expected to yield $\kappa \approx 0.55$–0.65 pre-calibration and $\kappa \approx 0.78$–0.85 post-calibration, consistent with published human annotation studies on complex NLP tasks.

4 Annotation Poisoning at the Step Level

4.1 The Pre-Gradient Intervention Point

Step-level annotation is the only point in the PRM training pipeline where a malicious actor can intervene *before* the gradient update. Once poisoned labels enter the training batch, the model's optimizer has no mechanism to distinguish a high-quality step from a maliciously mislabeled one. This makes annotation-layer integrity a safety-critical property, not merely a data quality concern.

The risk is quantified by the *gradient conflict rate*: the proportion of training steps that receive incorrect reward signals due to label poisoning. Consider an outcome-failed trajectory in which 14 of 15 steps were executed correctly, but the terminal step failed. Under ORM training, all 14 correct steps receive a -1 reward, producing gradient conflicts at every intermediate position. Under PRM training with poisoned step labels—where a malicious annotator assigns -1 to correct intermediate steps—the same gradient conflicts are produced, but they are now invisible to *any* task-success evaluation metric. Task-success metrics observe only the terminal outcome; they cannot distinguish correctly-labeled trajectories from poisoned ones.

We measure the gradient conflict rate empirically in AGENTTRACE. In our synthetic corpus, outcome-failed trajectories in which $\geq 50\%$ of intermediate steps were executed correctly exhibit a 100% gradient conflict rate under ORM reward assignment. This result is a synthetic artifact—in production, the rate depends on task difficulty distribution—but it establishes the upper bound of the problem and motivates step-level annotation as the correct intervention point. Souly et al. [2025] provide independent evidence for the severity of the risk: 250 maliciously labeled documents suffice to backdoor arbitrarily large models, regardless of clean data proportion.

4.2 MAD Detection Blind Spot

The standard defense against malicious annotators in large annotation pipelines is *Median Absolute Deviation* (MAD) suspicion scoring. Each annotator receives a suspicion score $s \in [0, 1]$ proportional to their deviation from the panel consensus on each annotation dimension. Annotators whose scores consistently deviate from the majority are flagged as outliers and removed.

This defense fails against a specific and practically relevant threat model: *coordinated collusion*, in which multiple annotators assign identical biased labels.

Formally, let $\hat{\mu}$ be the panel median for a given annotation dimension, and let $\delta_i = |x_i - \hat{\mu}|$ be the absolute deviation of annotator i. The MAD suspicion score for annotator i is:

$$s_i = \frac{\delta_i}{\text{MAD}(\delta_1, \ldots, \delta_n)} \tag{2}$$

where MAD $=$ median$(|\delta_i - \text{median}(\delta)|)$. When $k \geq 3$ annotators collude with identical biased labels, their joint deviation from the panel consensus is shared equally. Their mean deviation equals their individual deviations, so the MAD of the full panel shifts toward the colluders' shared bias. The colluding annotators' suspicion scores collapse toward $s = 0.0$— indistinguishable from the most reliable annotator in the pool.

We confirm this empirically by injecting three coordinated poisoners (Poisoner_A, Poisoner_B, Poisoner_C) into our five-annotator panel. All three poisoners receive suspicion scores of $s = 0.000$ at every detection threshold (Table 2). A threshold sweep from 0.0 to 1.0 yields $F_1 = 0.0$ throughout: precision and recall are both zero because no poisoner is ever flagged. This is not an implementation failure—it is a fundamental property of MAD-based detection. Any detection scheme that relies solely on deviation from panel consensus will share this blind spot when the colluding group is large enough to shift the consensus.

Table 2: MAD suspicion scores for the augmented annotator pool (5 legitimate + 3 injected colluding poisoners). Poisoners achieve the lowest suspicion scores in the pool, making threshold-based detection impossible. $F_1 = 0.0$ at all thresholds.

Annotator	Type	MAD Suspicion Score
RecoverySkeptic	Legitimate	1.000
OutcomeOptimist	Legitimate	0.683
ProcessPurist	Legitimate	0.488
ClinicalSafetyFirst	Legitimate	0.293
PrivacyMaximalist	Legitimate	0.195
Poisoner_A	Injected	0.000
Poisoner_B	Injected	0.000
Poisoner_C	Injected	0.000

4.3 Second-Layer Defense: Confident Learning

The MAD blind spot motivates a complementary defense that operates on *label distributions* rather than annotator identity. *Confident Learning* Northcutt et al. [2021] detects noisy labels by estimating the joint distribution of annotator-assigned labels and latent true labels, identifying examples where the assigned label is inconsistent with the model's confident predictions. Crucially, this approach is blind to annotator identity and collusion structure: it flags suspicious labels regardless of *who* assigned them.

We integrate Cleanlab's `find_label_issues` with quality score thresholding and `get_label_quality_scores` into AGENTTRACE as a second-layer defense. In the pilot study, Cleanlab identifies zero label issues in the clean annotation pool—confirming no false positives on legitimate annotators. In future work with real poisoned annotation data, Confident Learning is expected to surface the label-level signature of colluding poisoners even when their annotator-level scores are undetectable via MAD. The two defenses are complementary: MAD catches individual outlier annotators; Confident Learning catches systematic label corruption regardless of its source.

5 The AGENTTRACE Pipeline

5.1 Architecture

AGENTTRACE is a six-stage, end-to-end pipeline from synthetic wearable trajectory generation to annotation quality evaluation. Each stage produces structured output that feeds directly into the next, enabling full reproducibility from a single seed. Table 3 summarizes the stages, modules, inputs, outputs, and key metrics.

Table 3: AGENTTRACE pipeline summary. All results from `seed=42`; 100 synthetic trajectories; 5 annotator personas.

Stage	Module	Input	Key Metric
1. Synthetic generation	`wearable_generator.py`	seed=42	5 scenario types $\times$ 20 logs
2. Differential privacy	`privacy_gate.py`	Raw sensor data	$\varepsilon = 1.0$
3. Multi-persona annotation	`annotator_simulator.py`	100 logs, 5 personas	Pre-cal $\kappa = -0.035$
4. IAA calibration + PIA	`pia_calculator.py`	150 annotation records	κ: $-0.065 \rightarrow +0.743$
5. PRM step labeling	`prm_annotator.py`	20 trajectories	100% gradient conflict rate
6. Poisoning detection	`poisoning_detector.py`	25 records + 3 poisoners	MAD $F_1 = 0.0$; Cleanlab layer

Stage 1 — Synthetic trajectory generation. The pipeline generates 100 wearable agentic trajectories across five scenario types: medication adherence, continuous vitals monitoring, behavioral pattern detection, emergency response, and privacy-sensitive ambient capture. Each scenario type contributes 20 balanced trajectories, ensuring coverage of the full range of agent behaviors and failure modes relevant to ambient wearable AI.

Stage 2 — Differential privacy gate. Raw sensor values (heart rate, SpO_2, accelerometer readings) are passed through a differential privacy (DP) gate before annotation, adding calibrated Gaussian noise with privacy budget $\varepsilon = 1.0$. This ensures that annotators never observe exact biometric values, respecting contextual consent boundaries while preserving the annotation signal necessary for trajectory quality assessment.

Stages 3–4 — Multi-persona annotation and PIA scoring. Five annotator personas (OutcomeOptimist, RecoverySkeptic, ProcessPurist, ClinicalSafetyFirst, PrivacyMaximalist) annotate each trajectory on the four PIA rubric dimensions. Pre-calibration standard κ is -0.035 (poor); PIA scoring recovers $\kappa = +0.743$ (substantial) as reported in §3.

Stages 5–6 — PRM labeling and poisoning detection. Step-level process reward labels are assigned to 20 trajectories, producing 60 step reward records. The poisoning detection module applies MAD suspicion scoring followed by Cleanlab Confident Learning as described in §4.

5.2 Dataset

We release an annotated wearable agentic trajectory dataset on HuggingFace at `https://huggingface.co/datasets/finaspirant/wearable-agent-trajectory-annotations`. The dataset contains 100 synthetic wearable trajectories with full multi-persona PIA annotations, step-level process reward labels, and DP-gated sensor data. The annotation schema is organized in three layers:

- **Session level:** trajectory metadata, scenario type, agent framework, terminal outcome.
- **Role level:** per-annotator-persona dimension scores and calibration metadata.
- **Step level:** per-step process reward labels, tool invocation records, error flags, and privacy boundary assessments.

The dataset includes Croissant metadata Akhtar et al. [2024] for machine-readable dataset documentation, as required by the NeurIPS E&D track. The dataset comprises the following artifacts: 100 synthetic wearable trajectory logs across 5 scenario types (health alert, medication adherence,

vitals monitoring, behavioral pattern detection, and privacy-sensitive ambient capture); 150 multi-persona annotation records produced by 5 annotator personas across 30 trajectories on 4 rubric dimensions (planning quality, error recovery, goal alignment, privacy compliance); 60 step-level process reward labels across 20 trajectories; and 25 poisoning detection records including 3 injected adversarial annotations for reproducible blind-spot evaluation.

5.3 Reproducibility

The full pipeline is available at:

`https://github.com/finaspirant/llm-wearable-agentic-eval-pipeline`

All results reported in this paper are reproducible from `seed=42` using the `uv` environment manager. The repository includes executed Jupyter notebooks (`curation_pipeline_e2e_executed.ipynb`, `agentic_eval_flywheel_executed.ipynb`) with all output cells preserved, enabling inspection of intermediate results without re-running the full pipeline.

6 Evaluation: Curation Impact

6.1 Experimental Setup

To quantify the downstream value of annotation quality, we conduct an A/B experiment comparing trajectories produced by the full AGENTTRACE curation pipeline (*curated*, $n = 50$) against unfiltered raw trajectories (*raw*, $n = 50$) on six evaluation metrics. Raw trajectories bypass all annotation quality gates: no PIA calibration, no poisoning detection, and no PRM label validation. Curated trajectories pass through the full six-stage pipeline.

Both conditions are evaluated on six metrics inspired by the Kore.ai enterprise agent evaluation framework Kore.ai [2025]:

- **Trajectory success rate:** proportion of trajectories reaching the stated goal within the allowed step budget.
- **Tool invocation accuracy:** proportion of tool calls that match the expected tool, arguments, and invocation order.
- **Groundedness score** (RAGAS Es et al. [2024]): factual consistency of agent responses against a reference knowledge base.
- **Privacy leak detection:** proportion of trajectories where the agent discloses DP-gated biometric data outside authorized consent boundaries.
- **Orchestrator correctness:** proportion of multi-agent delegation decisions that route to the correct sub-agent.
- **Latency SLA compliance:** proportion of trajectories completing within the defined latency service-level agreement.

All experiments use `seed=42` for full reproducibility. We note that this experiment uses synthetic trajectories and LLM-simulated annotators; generalization to human annotators and real-world deployments requires future validation (see §7).

6.2 Results

Table 4 presents the curation impact across all six evaluation metrics.

Meaningful lifts (*). The two metrics showing meaningful improvement directly reflect annotation quality: trajectory success rate lifts from 0.12 to 0.33 ($+177.8\%$), and tool invocation accuracy lifts from 0.36 to 1.00 ($+177.8\%$). Both improvements trace directly to the PIA calibration and PRM step labeling stages: curated trajectories filter out those with poor inter-annotator agreement on tool precision (a PIA sub-dimension of goal alignment), retaining only trajectories where annotators agree on the correctness of each tool invocation. This quality gate is invisible to outcome-only evaluation—both the raw 0.36-accuracy trajectories and the curated 1.00-accuracy trajectories may reach the stated goal—but it directly drives downstream training signal quality under PRM training.

Table 4: A/B experiment: raw vs. curated trajectories ($n = 50$ per condition; `seed=42`). Starred metrics show statistically meaningful lifts; dagger metrics are bounded by synthetic data properties.

Metric	Raw	Curated	Δ (abs)	Δ (%)
Trajectory success rate⋆	0.12	0.33	+0.21	+177.8%
Tool invocation accuracy⋆	0.36	1.00	+0.64	+177.8%
Groundedness (RAGAS)†	0.75	0.75	+0.00	—
Privacy leak detection†	0.00	0.00	+0.00	—
Orchestrator correctness†	1.00	1.00	+0.00	—
Latency SLA compliance†	1.00	1.00	+0.00	—

⋆ Meaningful lift from annotation quality curation. † Bounded by synthetic data properties (see text).

Bounded metrics (†). Four metrics show zero delta between conditions. Groundedness ($\Delta = 0$, RAGAS score $= 0.75$ in both conditions) reflects a known RAGAS fallback behavior: when no retrieval context is provided, RAGAS returns a fixed fallback value of 0.75 rather than computing factual consistency. Privacy leak detection and orchestrator correctness reach floor/ceiling values of 0.00 and 1.00 respectively by synthetic data construction—the generator initializes all trajectories with correct orchestration and no privacy violations. Latency SLA compliance similarly reflects the deterministic latency of simulated agent calls. These zero-delta results are not failures of curation; they reflect synthetic data boundaries. Live evaluation with real agent API calls and a wearable knowledge base is expected to show curation lifts on all six metrics.

7 Discussion and Limitations

Domain specificity of PIA dimensions. The three core PIA rubric dimensions—planning quality, error recovery, and goal alignment—were designed for wearable agentic contexts. Other deployment domains may require different or additional dimensions. A customer service agent evaluation might prioritize tone consistency and escalation appropriateness over privacy compliance. A coding agent evaluation might prioritize test coverage and refactoring quality over error recovery. PIA is a methodology, not a fixed rubric: the core insight—evaluate rubric dimensions independently of path length—generalizes, even though the specific dimensions must be defined per domain. We provide the full rubric definition and calibration protocol in the repository to support adaptation to new domains.

Synthetic corpus and LLM annotator simulations. All results in this paper are derived from a synthetic trajectory corpus generated with `seed=42` and annotated by LLM-simulated personas, not human annotators. The LLM simulation produces deterministic annotation in dry-run mode, which yields the post-calibration $\kappa = 1.0$ artifact noted in §3.3. Generalization of PIA to human annotators and real-world deployment requires empirical validation. We report the pre-calibration numbers as the empirically meaningful results and provide all code and data necessary for researchers to replicate the experiment with real annotation panels.

Colluding annotator detection remains an open problem. The MAD blind spot we characterize (§4.2) is fundamental, not incidental. Any detection scheme based solely on deviation from panel consensus will fail when the colluding group is large enough to shift the consensus. Confident Learning provides complementary coverage at the label level, but it too can be evaded by sufficiently sophisticated poisoners who understand the label distribution model. Graph-based anomaly detection across annotation sessions—tracking annotator behavior over time rather than within a single batch—is a promising direction for future work.

Gradient conflict rate is a synthetic upper bound. The 100% gradient conflict rate we report (§4.1) is a synthetic artifact: the trajectory generator initializes all trajectories with `outcome_success=False`,

ensuring every trajectory is outcome-failed. In production, gradient conflict rates depend on the difficulty distribution of the task set and the agent's actual success rate. The finding remains useful as a proof of concept: it demonstrates the mechanism by which poisoned step labels propagate -1 reward to correct intermediate steps, and it establishes that step-level annotation is the correct intervention point.

8 Conclusion

We have identified two compounding failures in the standard evaluation stack for agentic AI systems. First, standard IAA metrics produce systematically misleading results for non-deterministic agents, misclassifying path divergence as annotator disagreement. Second, step-level annotation poisoning is invisible to task-success metrics and undetectable by standard MAD-based annotator screening when attackers coordinate.

PIA addresses the first failure by decoupling quality assessment from path length, recovering $\kappa = +0.743$ where standard metrics yield $\kappa = -0.065$. Confident Learning addresses the second as a complementary second-layer defense. AGENTTRACE operationalizes both in an open-source, reproducible pipeline, validated by an A/B experiment showing $+177.8\%$ lift in tool invocation accuracy and trajectory success rate from curated vs. raw trajectories.

We release the full pipeline, annotated dataset, and executed notebooks to enable reproducible research on agentic evaluation methodology—a prerequisite for the safe, reliable deployment of multi-step agentic AI systems in production.

Data Availability

All data, code, and artifacts introduced in this paper are publicly available:

- **Dataset (HuggingFace):** `https://huggingface.co/datasets/finaspirant/wearable-agent-trajectory-annotations` — 100 annotated wearable agentic trajectories with PIA scores, step-level PRM labels, and DP-gated sensor data. Includes Croissant metadata Akhtar et al. [2024].
- **Pipeline code (GitHub):** `https://github.com/finaspirant/llm-wearable-agentic-eval-pipeli` — Full six-stage AGENTTRACE pipeline. Reproducible from `seed=42` using the `uv` environment manager. Tagged release: `v1.0`.
- **Executed notebooks:** `notebooks/curation_pipeline_e2e_executed.ipynb` and `notebooks/agentic_eval_flywheel_executed.ipynb` are committed to the repository with all output cells preserved, enabling result inspection without re-running the full pipeline.
- **Demo (Loom):** `https://www.loom.com/share/5bec5764428b4e48aa868134e54a894e` — 5-minute walkthrough of the pipeline and key results.

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